

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import (
    confusion_matrix,
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    classification_report
)
```

```
df = pd.read_csv("WA_Fn-UseC_-Telco-Customer-Churn.csv")

print("Dataset loaded successfully!")

print("\nFirst 5 rows:")
print(df.head())

print("\nDataset Shape:")
print(df.shape)

print("\nColumn Names:")
print(df.columns.tolist())
```

Dataset loaded successfully!

First 5 rows:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	\
0	7590-VHVEG	Female	0	Yes	No	1	No	
1	5575-GNVDE	Male	0	No	No	34	Yes	
2	3668-QPYBK	Male	0	No	No	2	Yes	
3	7795-CFOCW	Male	0	No	No	45	No	
4	9237-HQITU	Female	0	No	No	2	Yes	

	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	\
0	No phone service	DSL	No	...	No	
1	No	DSL	Yes	...	Yes	
2	No	DSL	Yes	...	No	
3	No phone service	DSL	Yes	...	Yes	
4	No	Fiber optic	No	...	No	

	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	\
0	No	No	No	Month-to-month	Yes	
1	No	No	No	One year	No	
2	No	No	No	Month-to-month	Yes	
3	Yes	No	No	One year	No	
4	No	No	No	Month-to-month	Yes	

	PaymentMethod	MonthlyCharges	TotalCharges	Churn
0	Electronic check	29.85	29.85	No
1	Mailed check	56.95	1889.5	No
2	Mailed check	53.85	108.15	Yes
3	Bank transfer (automatic)	42.30	1840.75	No
4	Electronic check	70.70	151.65	Yes

[5 rows x 21 columns]

Dataset Shape:
(7043, 21)

Column Names:
['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure', 'PhoneService', 'MultipleLines', 'InternetService', 'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling', 'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn']

```
print("\nMissing Values:")
print(df.isnull().sum())

print("\nDuplicate Rows:")
print(df.duplicated().sum())
```

Missing Values:

customerID	0
gender	0

```

SeniorCitizen    0
Partner          0
Dependents       0
tenure           0
PhoneService     0
MultipleLines    0
InternetService  0
OnlineSecurity   0
OnlineBackup     0
DeviceProtection 0
TechSupport      0
StreamingTV      0
StreamingMovies  0
Contract         0
PaperlessBilling 0
PaymentMethod    0
MonthlyCharges   0
TotalCharges     0
Churn            0
dtype: int64

```

```

Duplicate Rows:
0

```

```
df = df.drop_duplicates()
```

```

df["TotalCharges"] = pd.to_numeric(
    df["TotalCharges"],
    errors="coerce"
)

# Fill missing TotalCharges with median

df["TotalCharges"] = df["TotalCharges"].fillna(
    df["TotalCharges"].median()
)

```

```
df = df.drop("customerID", axis=1)
```

```

df["Churn"] = df["Churn"].map({
    "Yes": 1,
    "No": 0
})

```

```

df = pd.get_dummies(
    df,
    drop_first=True
)

```

```

print("\nProcessed Dataset:")
print(df.head())

```

```

Processed Dataset:
  SeniorCitizen  tenure  MonthlyCharges  TotalCharges  Churn  gender_Male \
0              0        1           29.85         29.85      0         False
1              0       34           56.95        1889.50      0          True
2              0        2           53.85         108.15      1          True
3              0       45           42.30        1840.75      0          True
4              0        2           70.70         151.65      1         False

  Partner_Yes  Dependents_Yes  PhoneService_Yes \
0          True           False           False
1          False           False           True
2          False           False           True
3          False           False           False
4          False           False           True

  MultipleLines_No phone service ... StreamingTV_No internet service \
0              True ...                               False
1          False ...                               False
2          False ...                               False
3              True ...                               False
4          False ...                               False

  StreamingTV_Yes  StreamingMovies_No internet service  StreamingMovies_Yes \
0          False ...                               False
1          False ...                               False
2          False ...                               False

```

```

3          False          False          False
4          False          False          False

    Contract_One year  Contract_Two year  PaperlessBilling_Yes \
0          False          False          True
1          True          False          False
2          False          False          True
3          True          False          False
4          False          False          True

    PaymentMethod_Credit card (automatic)  PaymentMethod_Electronic check \
0          False          True
1          False          False
2          False          False
3          False          False
4          False          True

    PaymentMethod_Mailed check
0          False
1          True
2          True
3          False
4          False

[5 rows x 31 columns]

```

```

print("\nData Types:")
print(df.dtypes)

```

```

Data Types:
SeniorCitizen          int64
tenure                 int64
MonthlyCharges         float64
TotalCharges           float64
Churn                  int64
gender_Male            bool
Partner_Yes            bool
Dependents_Yes         bool
PhoneService_Yes       bool
MultipleLines_No phone service  bool
MultipleLines_Yes       bool
InternetService_Fiber optic    bool
InternetService_No         bool
OnlineSecurity_No internet service  bool
OnlineSecurity_Yes         bool
OnlineBackup_No internet service    bool
OnlineBackup_Yes           bool
DeviceProtection_No internet service  bool
DeviceProtection_Yes        bool
TechSupport_No internet service    bool
TechSupport_Yes             bool
StreamingTV_No internet service     bool
StreamingTV_Yes              bool
StreamingMovies_No internet service  bool
StreamingMovies_Yes           bool
Contract_One year           bool
Contract_Two year           bool
PaperlessBilling_Yes        bool
PaymentMethod_Credit card (automatic)  bool
PaymentMethod_Electronic check        bool
PaymentMethod_Mailed check            bool
dtype: object

```

```

X = df.drop("Churn", axis=1)

y = df["Churn"]

print("\nFeatures Shape:")
print(X.shape)

print("\nTarget Shape:")
print(y.shape)

```

```

Features Shape:
(7043, 30)

Target Shape:
(7043,)

```

```

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,

```

```
test_size=0.20,  
random_state=42,  
stratify=y  
)  
  
print("\nTraining Data:", X_train.shape)  
print("Testing Data:", X_test.shape)
```

Training Data: (5634, 30)
Testing Data: (1409, 30)

```
model = DecisionTreeClassifier(  
    criterion="gini",  
    max_depth=5,  
    random_state=42  
)
```

```
model.fit(  
    X_train,  
    y_train  
)  
  
print("\nDecision Tree Model Trained Successfully!")
```

Decision Tree Model Trained Successfully!

```
y_pred = model.predict(X_test)  
  
print("\nFirst 20 Predictions:")  
print(y_pred[:20])
```

First 20 Predictions:
[0 1 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 1 0 0]

```
accuracy = accuracy_score(  
    y_test,  
    y_pred  
)  
  
print("\nAccuracy:")  
print(round(accuracy, 4))
```

Accuracy:
0.7942

```
precision = precision_score(  
    y_test,  
    y_pred  
)  
  
print("\nPrecision:")  
print(round(precision, 4))
```

Precision:
0.6296

```
recall = recall_score(  
    y_test,  
    y_pred  
)  
  
print("\nRecall:")  
print(round(recall, 4))
```

Recall:
0.5455

```
f1 = f1_score(
    y_test,
    y_pred
)

print("\nF1 Score:")
print(round(f1, 4))
```

F1 Score:
0.5845

```
print("\nClassification Report:")
print(
    classification_report(
        y_test,
        y_pred
    )
)
```

Classification Report:

	precision	recall	f1-score	support
0	0.84	0.88	0.86	1035
1	0.63	0.55	0.58	374
accuracy			0.79	1409
macro avg	0.74	0.71	0.72	1409
weighted avg	0.79	0.79	0.79	1409

```
cm = confusion_matrix(
    y_test,
    y_pred
)

print("\nConfusion Matrix:")
print(cm)
```

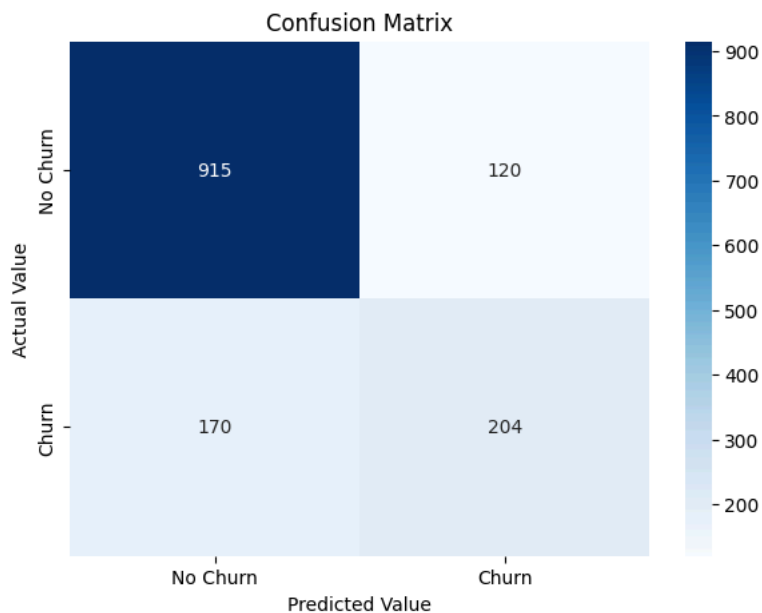
Confusion Matrix:
[[915 120]
[170 204]]

```
plt.figure(figsize=(7, 5))

sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=["No Churn", "Churn"],
    yticklabels=["No Churn", "Churn"]
)

plt.title("Confusion Matrix")
plt.xlabel("Predicted Value")
plt.ylabel("Actual Value")

plt.show()
```



```
feature_importance = pd.DataFrame({
    "Feature": X.columns,
    "Importance": model.feature_importances_
})

# Sort by importance

feature_importance = feature_importance.sort_values(
    by="Importance",
    ascending=False
)

print("\nFeature Importance:")
print(feature_importance)
```

```
Feature Importance:
```

	Feature	Importance
1	tenure	0.421435
10	InternetService_Fiber optic	0.357542
3	TotalCharges	0.047129
28	PaymentMethod_Electronic check	0.036761
9	MultipleLines_Yes	0.023630
12	OnlineSecurity_No internet service	0.020669
25	Contract_Two year	0.019599
22	StreamingMovies_No internet service	0.017379
2	MonthlyCharges	0.016847
19	TechSupport_Yes	0.012988
24	Contract_One year	0.008553
8	MultipleLines_No phone service	0.007049
13	OnlineSecurity_Yes	0.006628
27	PaymentMethod_Credit card (automatic)	0.002533
7	PhoneService_Yes	0.000975
0	SeniorCitizen	0.000283
4	gender_Male	0.000000
5	Partner_Yes	0.000000
11	InternetService_No	0.000000
6	Dependents_Yes	0.000000
16	DeviceProtection_No internet service	0.000000
14	OnlineBackup_No internet service	0.000000
21	StreamingTV_Yes	0.000000
20	StreamingTV_No internet service	0.000000
18	TechSupport_No internet service	0.000000
17	DeviceProtection_Yes	0.000000
15	OnlineBackup_Yes	0.000000
23	StreamingMovies_Yes	0.000000
26	PaperlessBilling_Yes	0.000000
29	PaymentMethod_Mailed check	0.000000

```
top_features = feature_importance.head(10)

print("\nTop 10 Important Features:")
print(top_features)
```

```
Top 10 Important Features:
```

	Feature	Importance
--	---------	------------

```

1          tenure      0.421435
10  InternetService_Fiber optic  0.357542
3          TotalCharges  0.047129
28  PaymentMethod_Electronic check  0.036761
9          MultipleLines_Yes  0.023630
12  OnlineSecurity_No internet service  0.020669
25          Contract_Two year  0.019599
22  StreamingMovies_No internet service  0.017379
2          MonthlyCharges  0.016847
19          TechSupport_Yes  0.012988

```

```

plt.figure(figsize=(10, 6))

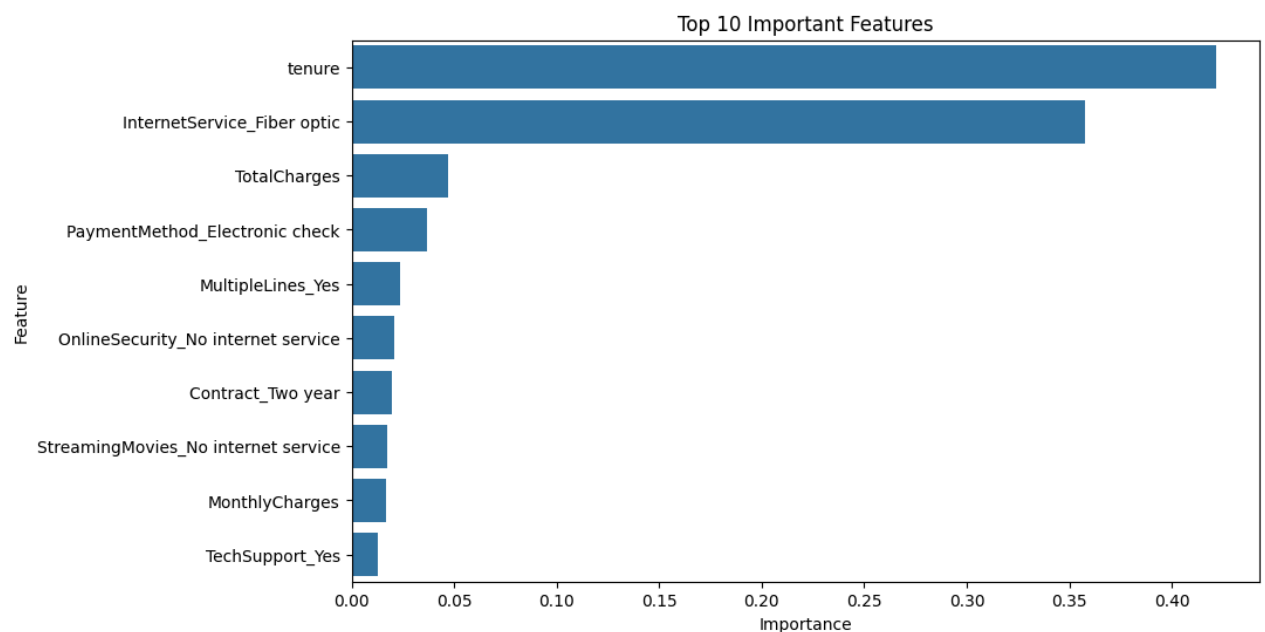
sns.barplot(
    data=top_features,
    x="Importance",
    y="Feature"
)

plt.title("Top 10 Important Features")

plt.xlabel("Importance")
plt.ylabel("Feature")

plt.show()

```



```

result = pd.DataFrame({
    "Actual Churn": y_test.values,
    "Predicted Churn": y_pred
})

print("\nActual vs Predicted:")
print(result.head(20))

```

```

Actual vs Predicted:
   Actual Churn  Predicted Churn
0             0             0
1             0             1
2             0             0
3             0             0
4             0             0
5             0             0
6             0             0
7             0             0
8             0             0
9             1             0
10            0             0
11            0             0
12            0             0
13            1             1
14            0             0
15            0             0

```

16	0	0
17	0	1
18	0	0
19	0	0

```
result.to_csv(
    "Churn_Prediction_Results.csv",
    index=False
)

feature_importance.to_csv(
    "Feature_Importance.csv",
    index=False
)

print("\nResults saved successfully!")
```

Results saved successfully!

```
print("\n=====")
print(" CUSTOMER CHURN PREDICTION COMPLETED")
print("=====")

print("Accuracy :", round(accuracy, 4))
print("Precision:", round(precision, 4))
print("Recall   :", round(recall, 4))
print("F1 Score :", round(f1, 4))
```

```
=====
CUSTOMER CHURN PREDICTION COMPLETED
=====
Accuracy : 0.7942
Precision: 0.6296
Recall   : 0.5455
F1 Score : 0.5845
```